Determine Surface Area of Prisms

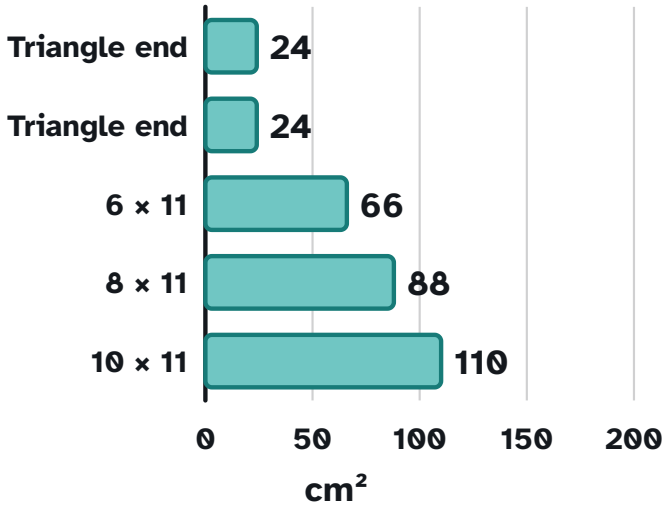
Find the surface area of a triangular prism by adding its two triangles and three rectangles.

Name _____ **Date** _____

Tonight's spark

A craft shop sells a tent-shaped wooden box. It is 11 cm long. Each triangle end is a right triangle with sides of 6, 8 and 10 cm. Paper covers every face. How many square centimeters of paper?

Guess first: is it more than 200 square centimeters, or less?



Our class model: the net with five faces

Try it

Work it out on paper. Write one answer on each line.

1 **A triangular prism has how many triangle ends and how many rectangles?**

A. 3 triangles and 2 rectangles **B.** 2 triangles and 3 rectangles
C. 2 triangles and 2 rectangles **D.** 1 triangle and 4 rectangles

My answer _____

2 **A triangle end of a prism has a base of 6 and a height of 4. What is its area?**

A. 12 square units **B.** 24 square units **C.** 10 square units

My answer _____

3 **A tent prism has triangle ends with sides 3, 4 and 5, and its length is 10. What is the area of the rectangle that uses the side of length 4?**

A. 14 **B.** 30 **C.** 50 **D.** 40

My answer _____

4 **The same tent has rectangles of 30, 40 and 50 square units, and each triangle end is 6 square units. What is the surface area?**

A. 126 square units **B.** 120 square units **C.** 132 square units **D.** 264 square units

My answer _____

Talk about it

Which faces use the length of the tent as one of their sides?

Home mission

Find a tent-shaped box, like a snack package or a toy tent. Count the faces out loud: triangles first, then rectangles. Which rectangle is the biggest?

If we get stuck, ask

- How many faces does this solid have? Name them one by one.
- Which side do all three rectangles share?
- Are we covering the outside or filling the inside?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-7-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it